

Juan David Muñoz-Bolaños

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Ms. Astrid Schernthaner

HR Business Partner
Besi Austria GmbH
Innstraße 16
6241 Radfeld, Austria

Subject: Application for Senior Vision Engineer (Job ID: 257595)

Dear Ms. Schernthaner,

I am writing to show my interest for the **Senior Vision Engineer** position at Besi Austria. As an R&D Optical Engineer currently living in **Innsbruck**, I offer a seamless transition into a full-time role. I have followed Besi's leadership in high-precision die-attach technology with interest, and the opportunity to drive the development of alignment systems in Radfeld aligns with my background in industrial vision and advanced metrology.

If I am not wrong, the position requires a candidate who can bridge the gap between "algorithmics" and "optics hardware." My career has been defined by this intersection:

1. Proven Industrial Vision Experience:

Before my M.Sc., I worked as a Machine Vision Engineer at INTECOL S.A.S. I successfully implemented **MVTec HALCON** solutions for high-speed bottling lines, designing custom illumination concepts to ensure robust **OCR** inspection. We implemented a safety-critical **laser profilometry** project for urban cable cars in Metro de Medellin. This experience ensures I can "create proofs of concept" that are reliable enough for production transfer.

2. Advanced Optics & Lab Development:

During my PhD in Innsbruck and my Master's in Jena, I specialized in the extreme precision required for semiconductor-scale inspection. I design and maintain complex optical labs featuring **Adaptive Optics** and **SLM/DMD** technology. I am currently integrating high-speed/precision Texas Instruments MEMS devices to correct optical aberrations, mirroring the high-speed alignment needs of modern die bonders.

3. Local & Legal Integration:

With a MINT degree from Germany and an upcoming Austrian PhD, I qualify for the **Red-White-Red Card (High-Skilled / Shortage Occupation)**. This ensures a straightforward hiring process, similar to EU candidate. Living in Innsbruck, I am already integrated into the Tyrolean region, with appreciation for the local quality of life: skiing, biking and dancing are my passions. I am committed to a long-term future in Tyrol.

I would be grateful to bring my "capacity for excellence" to Besi, translating my expertise in precision optics and industrial inspection into the next generation of chip placement machines. Thank you for considering my application.

Sincerely,

Juan David Muñoz-Bolaños

R&D Optical Engineer
Medical University of Innsbruck